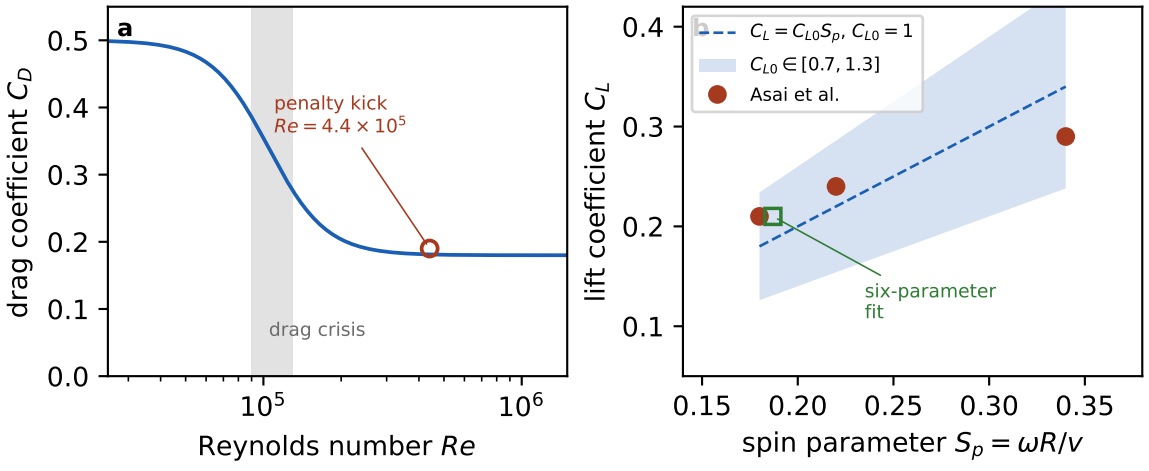
a
drag coefficient C_D
0.5
0.4
0.3
0.2
0.1
0.0
penalty kick
$Re = 4.4 \times 10^5$
drag crisis
10^5
10^6
Reynolds number Re
b
lift coefficient C_L
0.4
0.3
0.2
0.1
$C_L = C_{L0} S_p$, $C_{L0} = 1$
$C_{L0} \in [0.7, 1.3]$
Asai et al.
six-parameter
fit
0.15
0.20
0.25
0.30
0.35
spin parameter $S_p = \omega R / v$